

RNN-1: Multivariate Weather Forecasting

Data Mining — Deep Learning, Core Project (25 pts)

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Abstract

This report compares recurrent models on the Jena Climate series: given 72 hours of multivariate sensors, forecast the next 12 hours of temperature (°C) and relative humidity (percentage points). Persistence, a direct LSTM, a direct GRU, and an LSTM encoder-decoder are trained under one leakage-safe protocol. The original contribution is a horizon-specific additive-attention LSTM: each lead time has its own learned query over the 72 encoder states. The hypothesis was that this would help most at longer horizons by selecting recent trends and daily lags. On the held-out test split the opposite mostly happened. Direct GRU is the strongest model (test temperature MAE 0.55–1.78 °C from 1 to 12 h). Attention is worse at short leads and only catches up at 12 h, slightly winning humidity RMSE. All neural models beat persistence after 1 h. Errors grow with horizon and with sudden observed changes. Validation-residual 90% intervals on the selected GRU cover about 91–92% of test points.

1 Problem and data

The task is 12-hour multivariate forecasting from a sliding history [1, 2]. At origin t , the input is the previous $L = 72$ hourly feature vectors and the target is

$$(T_{t+1}, RH_{t+1}), \dots, (T_{t+12}, RH_{t+12}).$$

Temperature and humidity are kept in physical units and never averaged into one error.

Source. Jena Climate (Max Planck Institute for Biogeochemistry), downloaded with `keras.utils.get_file` from the official Keras ZIP [2]. Raw file: 420,124 ten-minute rows, 14 sensors, 1 Jan 2009 – 1 Jan 2017.

Cleaning. Duplicate timestamps (327) had identical values and were dropped, keeping the first. Wind-speed sentinels ≤ -9990 are treated as missing (none were present in this file). Wind direction is circular, so `wd (deg)` is replaced by sine/cosine before resampling; arithmetic averaging of degrees would mix 359° and 1° into 180° . Sensors are then resampled to hourly means (70,043 hours). Gaps are filled with causal forward-fill, at most six hours, then 86 unresolved rows are dropped. Cyclical hour-of-day and day-of-year features are added. The model sees 19 input channels: the numeric sensors (with wind encoded circularly) plus four time features.

Split and windows. A chronological 70/15/15 split, no shuffling of time:

Split	Period	Hours	Windows
Train	2009-01-01 – 2014-08-05	49,030	48,947
Validation	2014-08-05 – 2015-10-18	10,506	10,495
Test	2015-10-18 – 2017-01-01	10,507	10,496

Standardization is fit on training rows only and inverted before MAE/RMSE. For a target starting at index i , the input is $[i - 72, i)$ and the target is $[i, i + 12)$. Validation/test targets never cross a split boundary; their inputs may use the hours immediately before the boundary, which would be available at forecast time. Only training windows are shuffled. Seed 2026, TensorFlow 2.20, one GPU.

Figure 1 shows the cleaned targets and the split.

Jena Climate: 420,124 ten-minute rows $\rightarrow$ 70,043 hourly rows

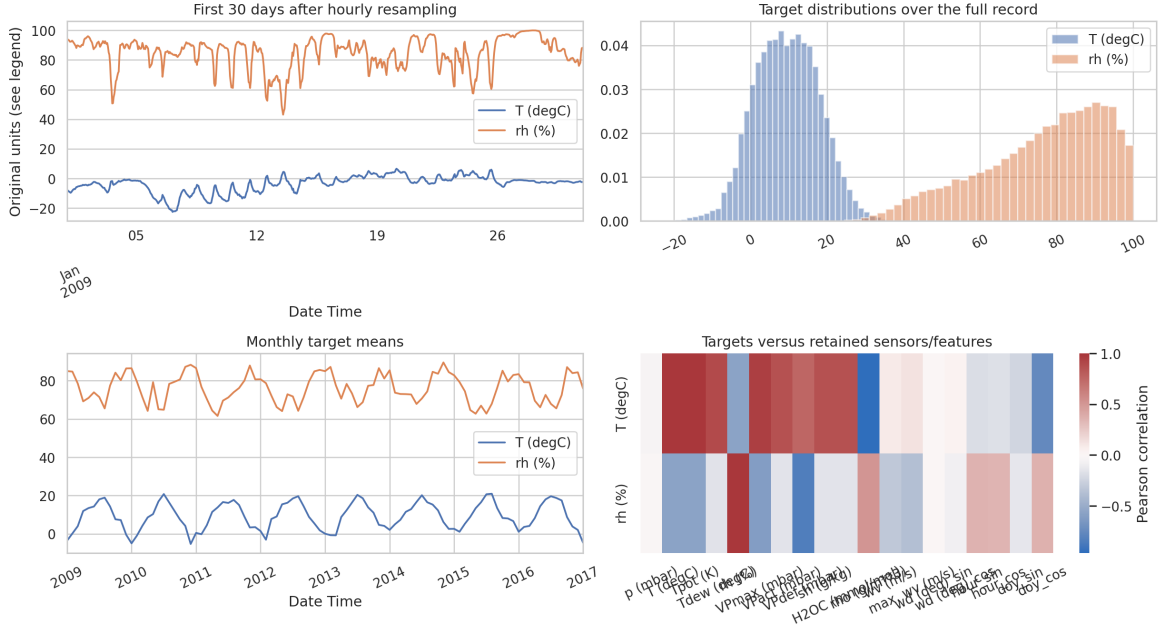


Figure 1: Hourly temperature and humidity, distributions, and chronological train/validation/test cuts.

2 Method

All neural models share width 64, dropout 0.1, Adam at 10^{-3} (halved on plateau), batch 128, standardized MSE, at most 30 epochs, and early stopping on validation loss with patience 6 and restore-best. They differ only in the recurrent cell and how the 12-step output is formed.

- **Persistence.** Repeat the last observed T and RH for every horizon. No parameters.
- **Direct LSTM / GRU.** One recurrent summary vector, then a dense head reshaped to 12×2 .
- **Seq2Seq LSTM.** Encoder LSTM produces context and state; the context is repeated 12 times and decoded by a second LSTM (no teacher forcing).
- **Horizon-attention LSTM.** Encoder LSTM returns all 72 states. This is the original model (Section 3).

Metrics. Per target and horizon: MAE and RMSE in $^{\circ}\text{C}$ or humidity percentage points. For a single ranking score we use *normalized aggregate MAE* (NMAE): each MAE divided by that target’s training standard deviation, then averaged. NMAE is dimensionless and is not a physical-unit error. Model selection for the interval study uses validation NMAE only.

3 Original contribution

A stock LSTM compresses 72 hours into one vector. The required novelty [1] is a **horizon-specific additive attention** over encoder states. Horizon h has its own embedding $\mathbf{q}_h$. That embedding is fused with the final LSTM state into a query; additive attention then produces weights $\alpha_{h,\ell}$ over look-back positions $\ell = 1, \dots, 72$ and a context $\mathbf{c}_h$. The forecast for lead h is read from $[\mathbf{c}_h; \mathbf{h}_{72}]$.

Hypothesis (stated before training). Short-horizon queries should concentrate on the most recent hours (local level and slope). Longer-horizon queries should put more mass near lags 24, 48, and 72 h, i.e. yesterday at the same time of day. If that is useful, attention’s gap versus direct LSTM/GRU should *widen* at 6–12 h, not at 1 h, where persistence and the last recurrent state are already strong.

Attention weights are associations inside this predictor, not a causal explanation of the weather.

4 Experiments

One full-mode Colab run (not FAST): stride 1, 64 units, seed 2026. Table 1 is the shared training log. Direct GRU trained longest (16 epochs) and reached the lowest scaled validation loss. Attention has more parameters than the direct models (47k vs. 18–23k) but fewer than Seq2Seq (55k), so a win for attention would not automatically be “more capacity.”

Table 1: Training under the shared protocol. Best validation loss is standardized MSE.

Model	Params	Epochs	Time (s)	Best val. loss
Direct LSTM	23,064	10	61	0.151
Direct GRU	17,880	16	82	0.143
Seq2Seq LSTM	54,658	10	69	0.147
Horizon Attention LSTM	47,170	10	81	0.150

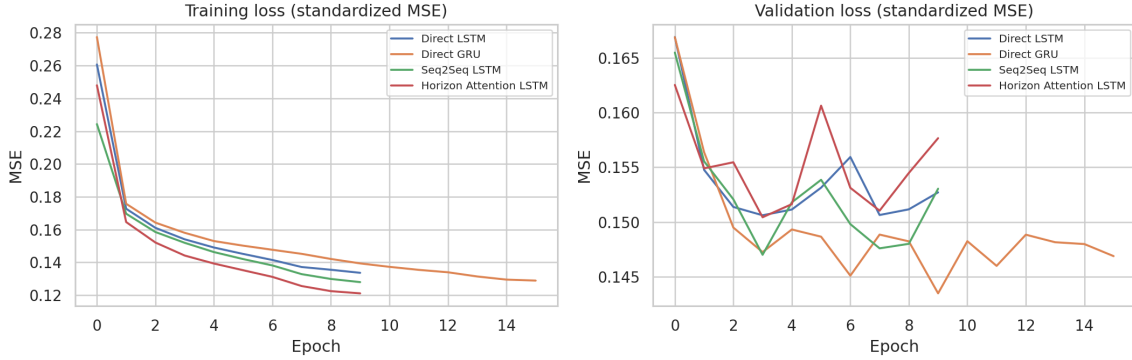


Figure 2: Train and validation standardized MSE. Checkpoints restore the best validation epoch.

Validation NMAE selected **Direct GRU** (0.259), then Seq2Seq (0.264), Direct LSTM (0.267), Attention (0.270). Test metrics were not used for that choice.

5 Results and analysis

Tables 2–3 report test MAE/RMSE at 1, 3, 6, and 12 hours. Figure 3 shows all 12 leads. Three facts are stable across both targets.

All recurrent models beat persistence after 1 h. At 1 h, GRU is best (temperature MAE 0.553 °C vs. persistence 0.643). Direct LSTM is slightly worse than persistence at 1 h temperature (0.662). From 3 h onward the gap is large: persistence 12 h temperature MAE is 4.19 °C against 1.78–1.84 for the neural models.

Error compounds with horizon. GRU temperature MAE goes 0.55 → 0.92 → 1.33 → 1.78 °C; humidity MAE 2.32 → 4.25 → 5.82 → 7.29 pp. Direct models have no autoregressive feedback, so this is mostly intrinsic unpredictability, not exposure bias. Seq2Seq could in principle feed its own errors forward; here it tracks the direct models closely, so that extra pathway did not dominate.

The attention hypothesis is mostly rejected. Attention is the worst neural model at 1–6 h on both targets (e.g. 1 h temperature MAE 0.833 vs. GRU 0.553). By 12 h it has almost closed the gap: temperature MAE 1.806 vs. GRU 1.783, and it is slightly best on 12 h humidity MAE (7.260) and RMSE (9.388). Relative to Direct LSTM it does not pull away at long leads in a way that would support the hypothesis. Extra parameters and a harder optimization problem are enough to explain a weak short-horizon model that later catches up. Test NMAE ranks GRU 0.243, Seq2Seq 0.250, LSTM 0.252, Attention 0.255, persistence 0.514.

Table 2: Test temperature errors. Bold is best in column.

Model	MAE (°C)				RMSE (°C)			
	1 h	3 h	6 h	12 h	1 h	3 h	6 h	12 h
Persistence	0.643	1.721	2.997	4.188	0.901	2.352	3.990	5.407
Direct LSTM	0.662	1.004	1.388	1.836	0.861	1.315	1.797	2.340
Direct GRU	0.553	0.917	1.329	1.783	0.725	1.220	1.717	2.286
Seq2Seq LSTM	0.643	0.994	1.388	1.842	0.839	1.318	1.812	2.365
Horizon Attn. LSTM	0.833	1.096	1.409	1.806	1.056	1.413	1.808	2.288

Table 3: Test relative-humidity errors (percentage points). Bold is best in column.

Model	MAE (pp)				RMSE (pp)			
	1 h	3 h	6 h	12 h	1 h	3 h	6 h	12 h
Persistence	2.773	7.070	11.938	16.442	3.976	9.609	15.654	20.848
Direct LSTM	2.451	4.376	5.936	7.393	3.351	5.994	7.896	9.574
Direct GRU	2.323	4.248	5.817	7.292	3.171	5.861	7.797	9.537
Seq2Seq LSTM	2.766	4.363	5.886	7.266	3.765	5.986	7.890	9.482
Horizon Attn. LSTM	3.066	4.526	5.937	7.260	4.138	6.241	7.944	9.388

Prediction examples. Figure 4 shows 12-hour rollouts for the validation-selected GRU on a fixed test segment and typical/hard windows. Smooth diurnal swings are captured; sharp turning points are lagged or damped.

Attention maps. Figure 5 is the diagnostic the hypothesis asked for. Average test attention and expected look-back by horizon, plus mass near 24/48/72 h lags (± 1 h). If the story were true, expected lag and daily mass should rise with horizon. The plots should be read against Tables 2–3: even a visually plausible daily peak does not make attention the better forecaster. The high-error heatmap is one worst-case window, not a typical day.

Sudden changes and failures. A change score is the maximum absolute one-hour jump over the 12-hour target path, including the step from the last observed hour. The 90th percentile of a combined (std-normalized) score is fixed on validation, then applied to test. Figure 6 plots GRU error against that score: errors rise when the observed weather moves quickly. Failure cases are the three highest-error windows among the most abrupt 10% of test samples — not hand-picked. Typical failure is lag and underestimation of amplitude, not a wild oscillation.

Stretch: 90% residual intervals. After freezing Direct GRU, per-target per-horizon absolute validation residuals give a conformal-style 90% quantile. Table 4 reports test coverage and mean width. Coverage is slightly conservative (about 91–92%). Width grows with horizon (2.4 $\rightarrow$ 8.0 °C; 10 $\rightarrow$ 33 humidity points), which matches the point-forecast error curves. These are symmetric marginal intervals. They

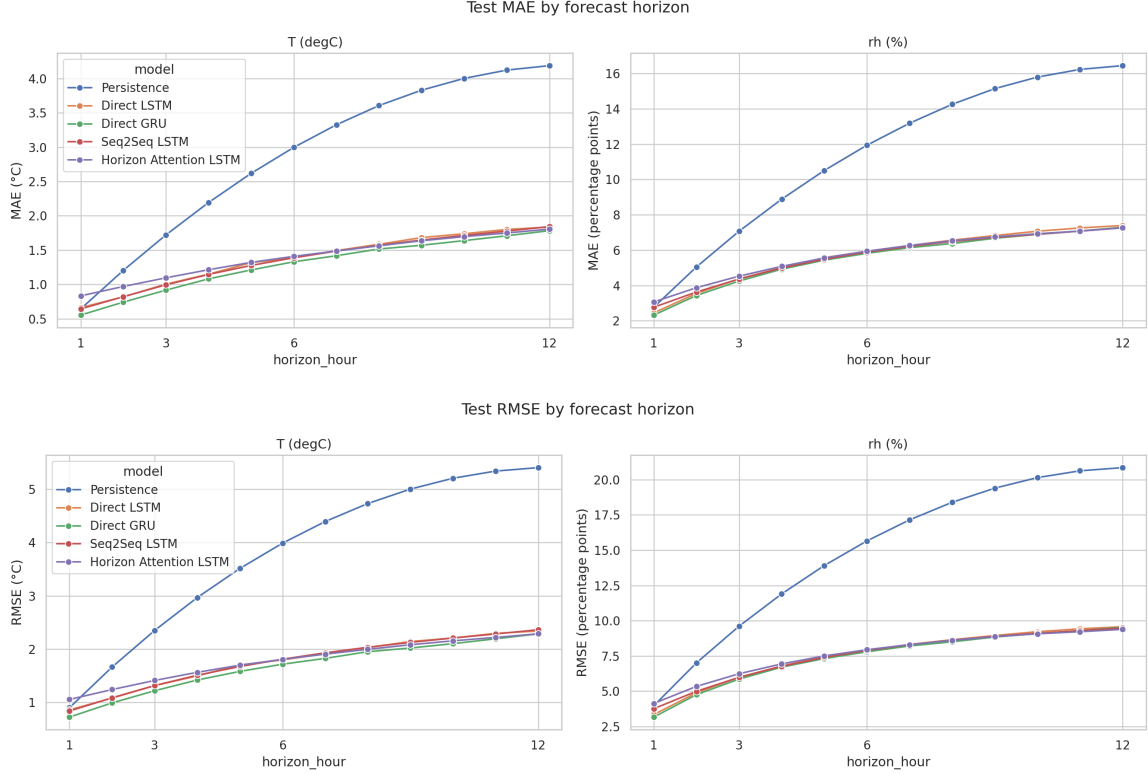


Figure 3: Test MAE and RMSE versus horizon. Separate panels keep °C and humidity points on their own scales.

do not give joint coverage over 24 outputs, do not adapt to regime, and are not guaranteed under serial dependence.

Table 4: Direct GRU, 90% validation-residual intervals on test ($n_{\text{cal}} = 10,495$).

Target	Coverage				Mean width	
	1 h	3 h	6 h	12 h	1 h	12 h
Temperature (°C)	0.908	0.918	0.918	0.917	2.37	7.98
Humidity (pp)	0.904	0.913	0.913	0.916	10.18	33.08

6 Discussion and observation

The two questions below are from the project card [1].

Question 1

Attention was introduced partly to fix what recurrence struggles with. If you added an attention mechanism over past timesteps to your forecaster, what would it learn to “look back” at in a weather signal, and would you expect the gain to be larger at short or long horizons?

Answer. Recurrence has a bottleneck: 72 hours are folded into one state. Attention lets each forecast lead re-read the encoder states. In weather, the useful look-backs are not mysterious. At 1 h the last observation (and maybe the last few hours of slope) is almost sufficient — that is why persistence MAE is

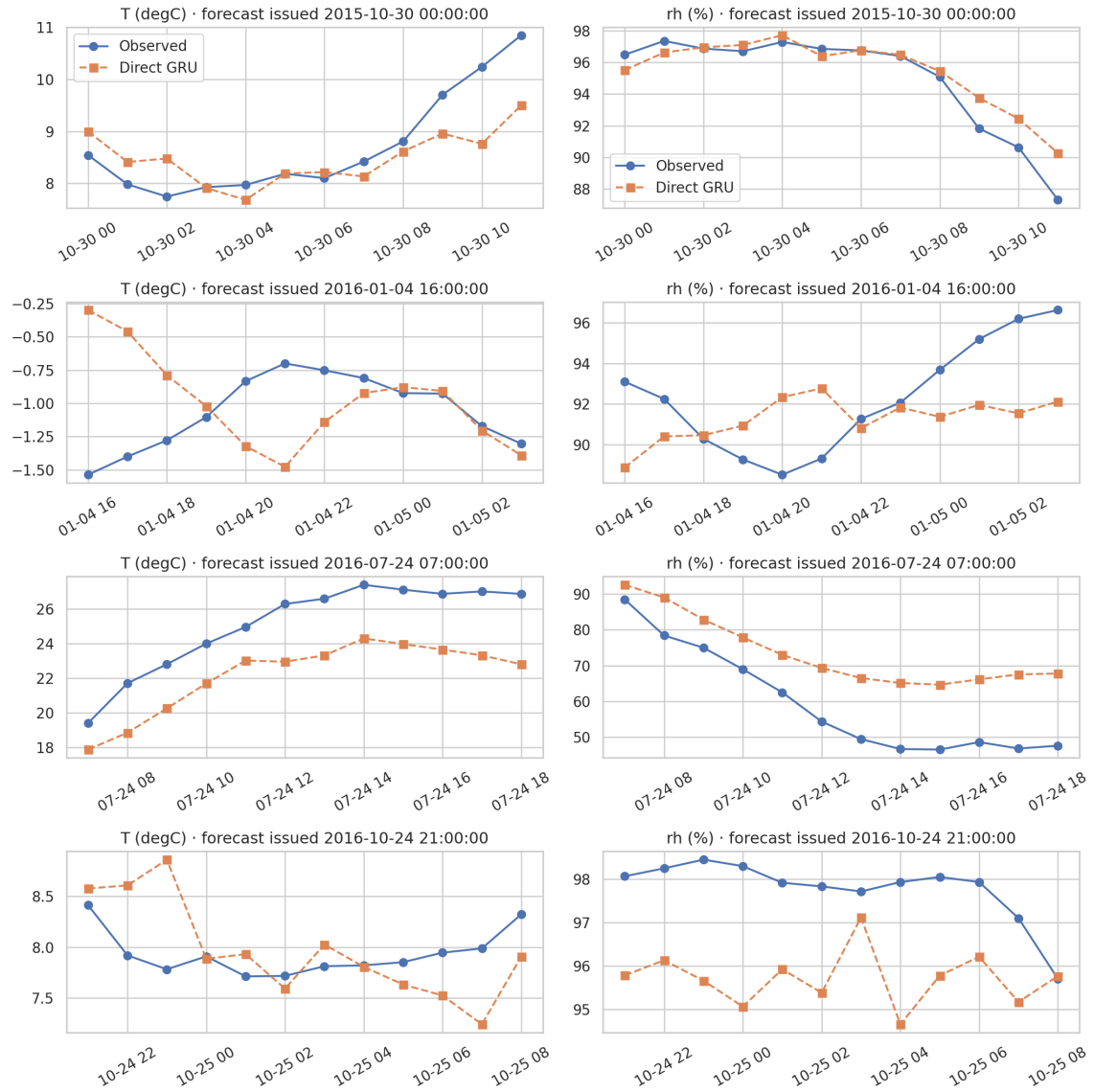


Figure 4: Observed vs. predicted 12-hour trajectories (Direct GRU). Temperature and humidity in separate panels.

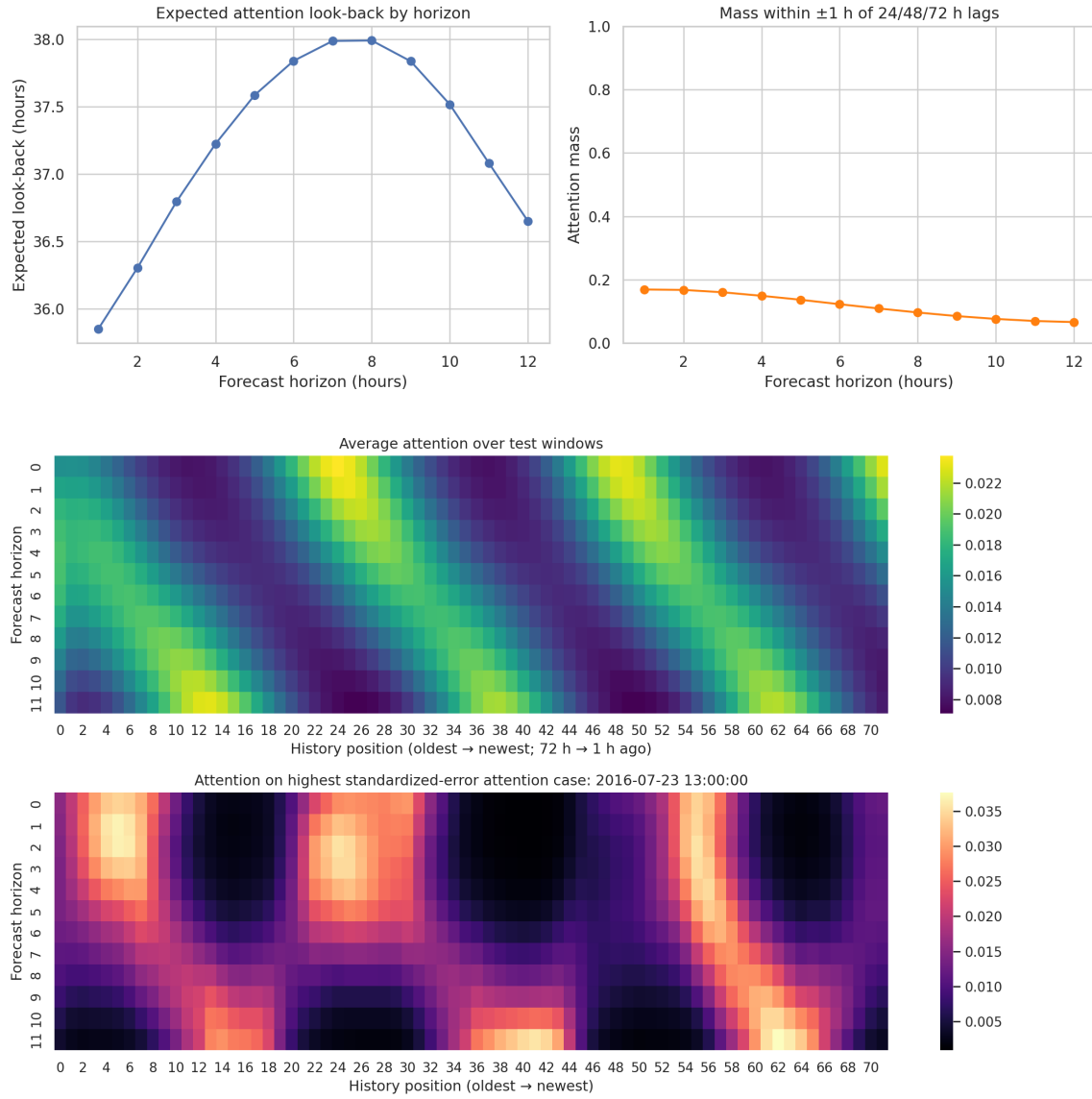


Figure 5: Top: expected attention lag and daily-lag mass vs. horizon. Bottom: mean test heatmap and the highest-error test window. Columns run from 72 h ago (left) to 1 h ago (right).

Residual magnitude vs sudden change — Direct GRU

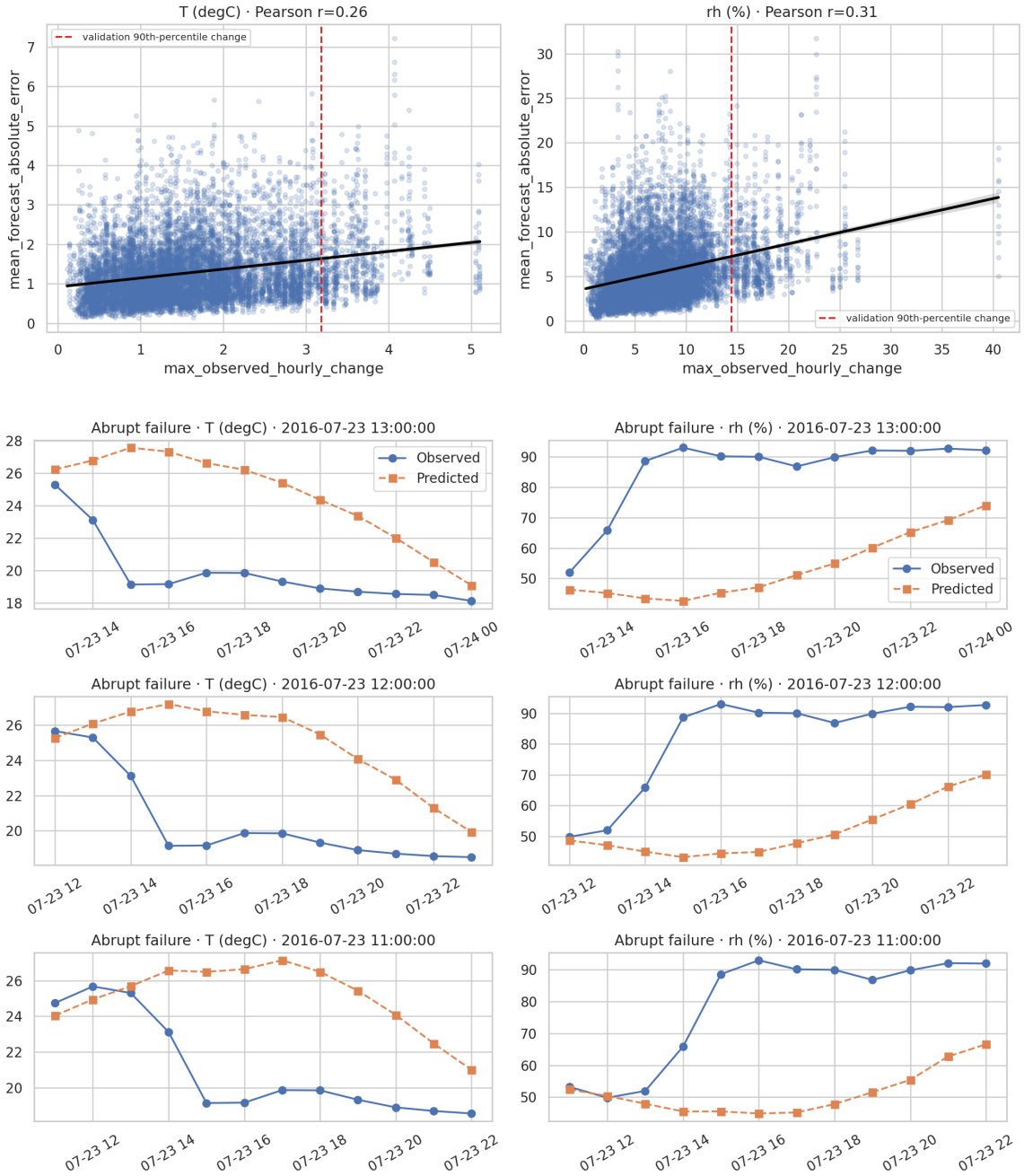


Figure 6: Top: GRU absolute error vs. observed hourly change (red line: validation 90th percentile). Bottom: three abrupt high-error test windows.

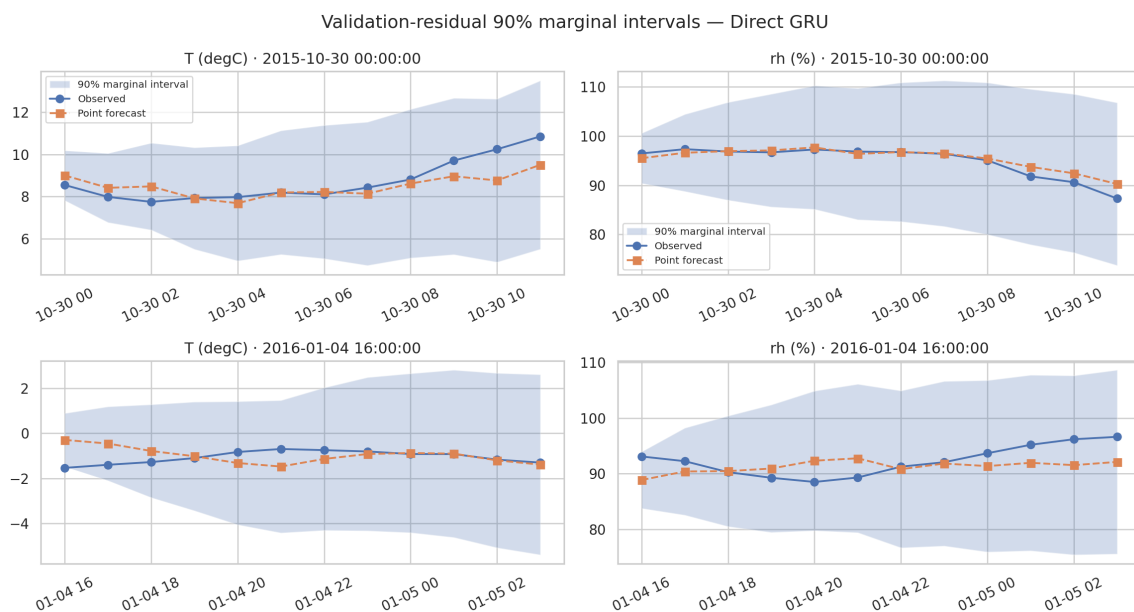


Figure 7: Example 90% residual intervals on the test period (Direct GRU).

already 0.64 °C. At 6–12 h, the same time yesterday and the day before (24/48/72 h) is a natural diurnal prior, and pressure/wind/humidity states can mark whether a pattern is repeating or breaking.

I therefore expected the *gain* to be larger at long horizons, and almost zero (or negative) at 1 h, where extra parameters mainly add variance. That directional prediction was only half right. Attention *was* worse at 1–3 h, and it did relatively better at 12 h (even winning humidity RMSE). It did not, however, beat a well-tuned direct GRU at long leads on temperature, and it lost overall NMAE. So “attention should look at yesterday” is a reasonable inductive story; it is not, on this single-station 72-step task, a better forecaster than a smaller GRU that just maps the last hidden state to all 12 hours. The heatmaps can look structured and still not buy accuracy — that is the honest result, not a plot I would hide.

Question 2

A Transformer can forecast the same series with no recurrence at all. What does the RNN’s sequential inductive bias buy you that a Transformer lacks here, and in what regime would you still bet on the RNN?

Answer. An LSTM/GRU applies the same local transition at every hour and carries a compact state. That bias matches a slowly evolving physical series: nearby hours are more related than distant ones, and you can update the forecast when a new hour arrives without recomputing attention over the whole window. Parameter sharing across time is also data-efficient on one station and $\sim 49k$ training windows.

A Transformer has no built-in left-to-right dynamics. Order has to be injected with positional or calendar encodings, which we already have as features. Its advantage is direct access between any pair of hours and parallel training. At $L = 72$ the $O(L^2)$ cost is not the issue; capacity and tuning are. This notebook does not train a Transformer, so I am not claiming RNNs won that comparison.

I would still bet on the RNN in the regime of this project: medium length (tens to a few hundred steps), modest data from one series, streaming updates, and a small compute budget. I would switch to a Transformer if the look-back became multi-week, if many stations were trained together, or if a pretrained time-series backbone were available. The sequential bias is a prior, not a theorem that recurrence must beat attention — my own attention model lost to a GRU.

7 Limitations

- One seed, one chronological split, one station. Overlapping windows are serially dependent; seasonal shift between 2009–2014 train and 2015–2016 test is possible.
- Hourly means remove sub-hour extremes. Causal fill of up to six hours can delay a real jump.
- Architectures are not parameter-matched. GRU is smallest and won; that confounds “GRU vs. attention” with capacity and optimization.
- Persistence is the only non-neural baseline. Seasonal persistence or a linear model would make the 1 h comparison sharper.
- Attention is not causal. Residual intervals are marginal and symmetric.

8 Conclusion

A leakage-safe 72→12 hour Jena forecast is well within reach of a small GRU: test MAE about 0.55 °C / 2.3 pp at 1 h and 1.78 °C / 7.3 pp at 12 h, far better than persistence once the lead grows. The original horizon-specific attention model was a fair test of a stated hypothesis and did not win. It was weak at short horizons, only competitive at 12 h, and slightly best on long-horizon humidity RMSE. Errors grow with lead time and with sudden observed changes. For this dataset the simplest justified model is the direct GRU; attention is an interpretable extra, not an accuracy upgrade.

References

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